

Design & Analysis of Algorithms Lab [L-T-P: 0-0-3 Credits: 2]

Prerequisites: Design & Analysis of Algorithms

Category: OLD (Modified)

Syllabus :

Arrays: Implementation of array operations Stacks and Queues, Circular Queues: Adding, deleting elements **[6 Lectures]**

Merging Problem: Evaluation of expressions, operations on multiple stacks and queues.**[4 Lectures]**

Linked lists: Implementation of linked lists, inserting, deleting, and inverting a linked list. Implementation of stacks and queues using linked lists. Polynomial addition and multiplication. Sparse Matrix multiplication and addition. **[8 Lectures]**

Trees: Recursive and non-recursive traversal of trees; AVL tree implementation. **[6 Lectures]**

Hashing: Hash 12 table implementation, searching, inserting and deleting Searching and sorting techniques **[6 Lectures]**

Heaps

Graphs: Shortest Path Algorithms. Minimum Cost Spanning Tree, Graph/Tree traversals, All pair shortest paths **[6 Lectures]**

Sorting: Quick Sort, Insertion sort, Merge sort, **[4 Lectures]**

References:

1. T. A. Standish: Data Structures, Algorithms and Software Principles, Addison-Wesley, Reading, Mass., 1995.
2. L. Nyhoff, C++— An Introduction to Data Structures, Prentice Hall, Englewood Cliffs, 1998.
3. A. M. Tannenbaum and M. J. Augesestein: Data Structures Using PASCAL, Prentice Hall, New Jersey, 1981.
4. D. E. Knuth: The Art of Computer Programming. Vol. 1, 2nd ed. Narosa/Addison-Wesley, New Delhi/London, 1973.
5. T. A. Standish: Data Structure Techniques, Addison-Wesley, Reading, Mass., 1980.
6. E. Horowitz and S. Sahni: Fundamentals of Data Structures, CBS, New Delhi, 1977.

7. R. L. Kruse: Data Structures and Program Design in C, Prentice Hall of India, New Delhi, 1996.
8. A. Aho, J. Hopcroft, and J. Ullman: Data Structures and Algorithms, Addison-Wesley, Reading, Mass., 1983.
9. B. Salzberg: File Structures: An Analytical Approach, Prentice Hall, New Jersey, 1988.
10. T. Harbron: File System Structure and Algorithms, Prentice Hall, New Jersey, 1987.

Teachers: Prof. S. Mukhopadhyay and Prof. R. Dutta